

4. SYSTEM ANALYSIS AND CLASSIFICATION

Objective

This experiment is divided into four simulations:

1. Discrete-Time System
2. Causal/Non-Causal System
3. Analysis of Discrete System
4. Analysis of Continuous-Time System

Experiment 4.1 Discrete -Time System

Program:

```
% Discrete-Time System  
clc;  
clear;  
close all;  
% Input sequence  
n = 0:10;  
x = [1 2 3 4 5 6 7 8 9 10 11];  
% Discrete-Time System  
y = filter([1 1],1,x);  
  
% Display output  
disp('Output Sequence:');  
disp(y);
```

```

% Plotting
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(n,y,'filled');
grid on;
title('Output Sequence y(n)=x(n)+x(n-1)');
xlabel('n');
ylabel('Amplitude');

```

Expected Output:

When you run the MATLAB code, you should observe two subplots:

1. Input Sequence (Unit Step Sequence):

- o Characteristics: The first subplot shows a unit step sequence, where the amplitude is 1 for all $n \geq 0$ and 0 for $n < 0$. Since the sequence starts at $n=0$, all values are 1.

- o Description: This sequence is used as the input to the discrete-time system.

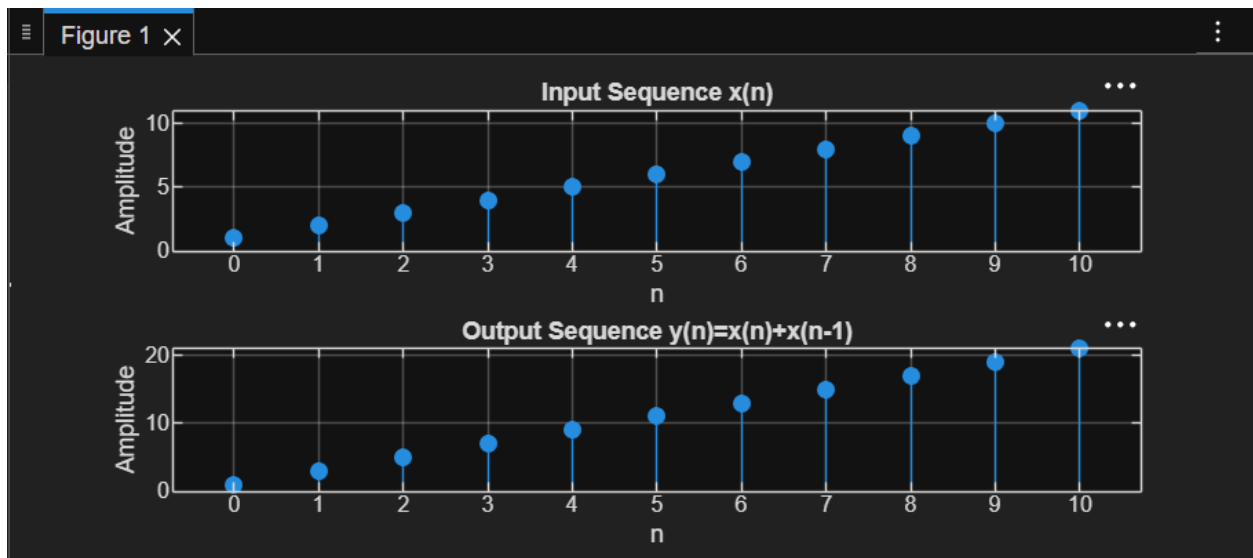
2. Output Sequence (System Response):

- o Characteristics: The second subplot shows the output sequence, which is the

system's response to the unit step input. The sequence should exhibit an increasing pattern that gradually approaches a steady state due to the recursive nature of the difference equation.

o Description: The output starts at 1 and increases by a fraction of the previous output value, eventually stabilizing as n increases.

Output Waveform:



Result:

The discrete-time system $y(n)=x(n)+x(n-1)$ was implemented using MATLAB. The output sequence was obtained by combining the present and previous input samples, and the input and output signals were displayed using stem plots.

Experiment 4.2 Causal/Non-Causal System

Program:

```
% Causal and Non-Causal Systems

clc;

clear;

close all;

% Input sequence
n = 0:10;
x = [1 2 3 4 5 6 7 8 9 10 11];

% Causal System:  $y(n)=x(n)+x(n-1)$ 
y_causal = filter([1 1],1,x);

% Non-Causal System:  $y(n)=x(n)+x(n+1)$ 
y_noncausal = x + [x(2:end) 0];

% Plotting

figure;

subplot(3,1,1);

stem(n,x,'filled');

grid on;

title('Input Sequence x(n)');

xlabel('n');

ylabel('Amplitude');

subplot(3,1,2);

stem(n,y_causal,'filled');

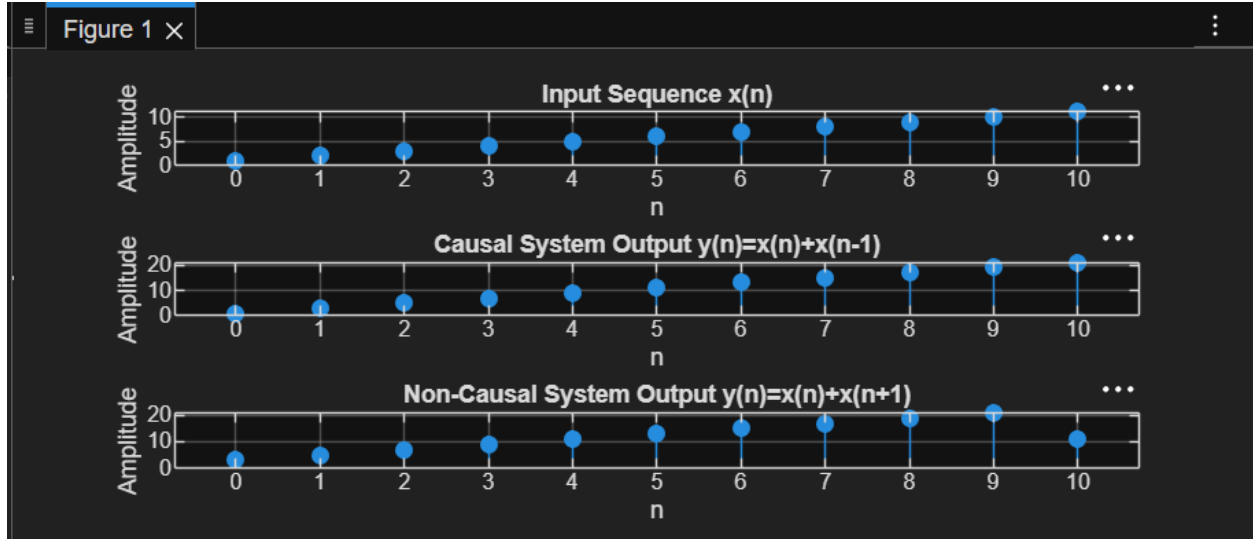
grid on;
```

```

title('Causal System Output  $y(n)=x(n)+x(n-1)$ ');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,3);
stem(n,y_noncausal,'filled');
grid on;
title('Non-Causal System Output  $y(n)=x(n)+x(n+1)$ ');
xlabel('n');
ylabel('Amplitude');

```

Output Waveform:



Result:

The causal and non-causal discrete-time systems were implemented successfully using MATLAB. The causal system used present and past input samples, whereas

the non-causal system used present and future input samples. The corresponding outputs were obtained and displayed using stem plots.